

CBSE BOARD (CLASS 10TH & 12TH) — SCIENCE QUESTION PAPER

Maximum Marks: 80 • Time Allowed: 3 Hours • Series: BOARD-2024 • Code: 31/1/1

General Instructions:

- This question paper consists of 39 questions in 5 sections.
- Section A consists of 20 objective type questions (1 mark each).
- Section B consists of 6 Very Short questions (2 marks each).
- Section C consists of 7 Short Answer type questions (3 marks each).
- Section D consists of 3 Long Answer type questions (5 marks each).
- Section E consists of 3 Case-Based source assessment units (4 marks each).

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each)

Q.1: Which of the following is a displacement reaction?

- (A) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$,
(B) $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$,
(C) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$,
(D) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$,

SECTION B: VERY SHORT ANSWER (2 Marks Each)

Q.21: State Ohm's Law and write the mathematical relationship between V, I, and R. [2 Marks]

SECTION E: CASE-BASED ASSESSMENT (4 Marks Each)

Q.37: Read the passage on Refraction of Light through a Glass Prism and answer sub-questions:

- (a) What causes the dispersion of white light? [1 Mark]
(b) Which colour of light deviates the most and why? [1 Mark]
(c) State Snell's law of refraction. [2 Marks]